



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

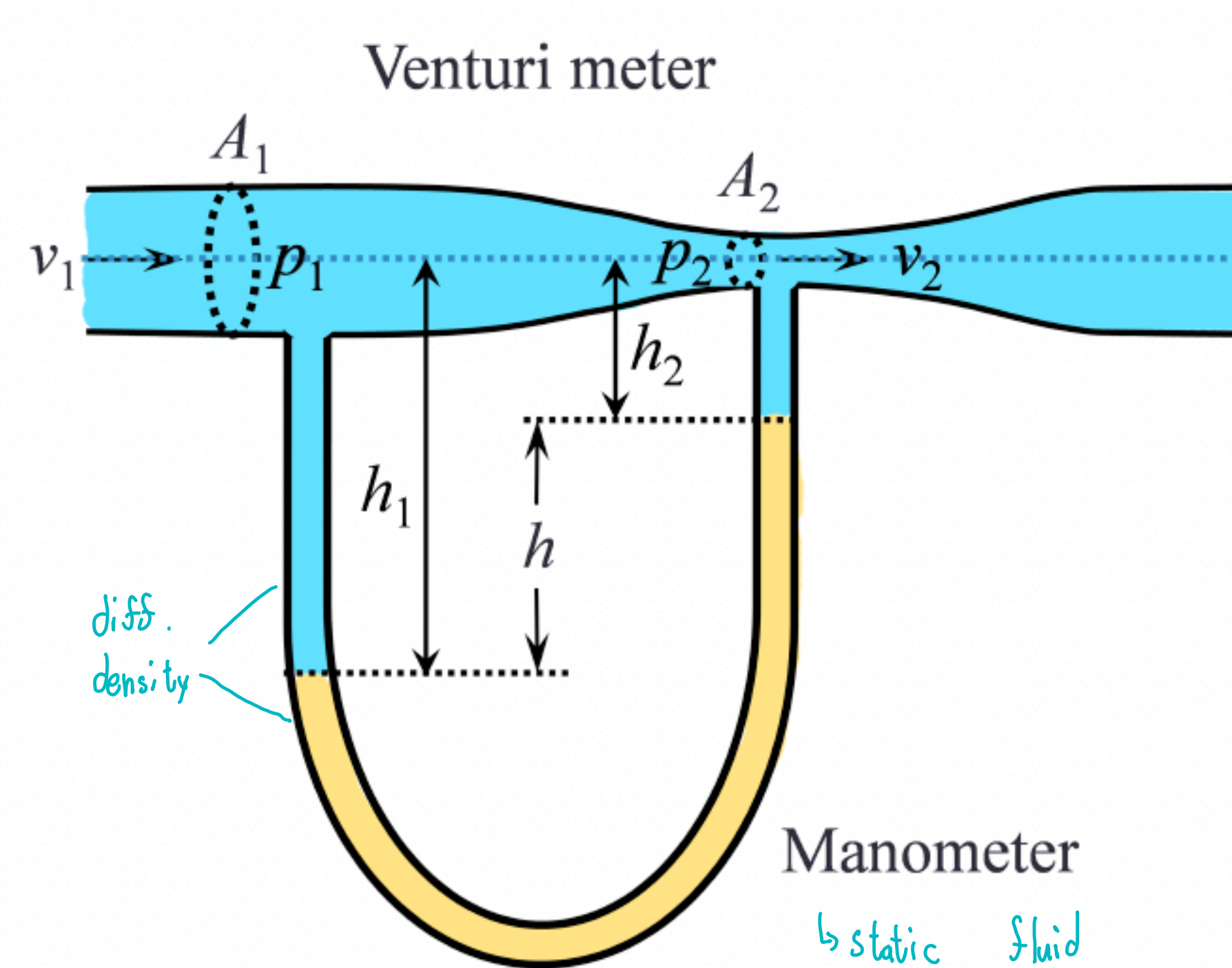
Lecture 22: Fluid, Oscillation

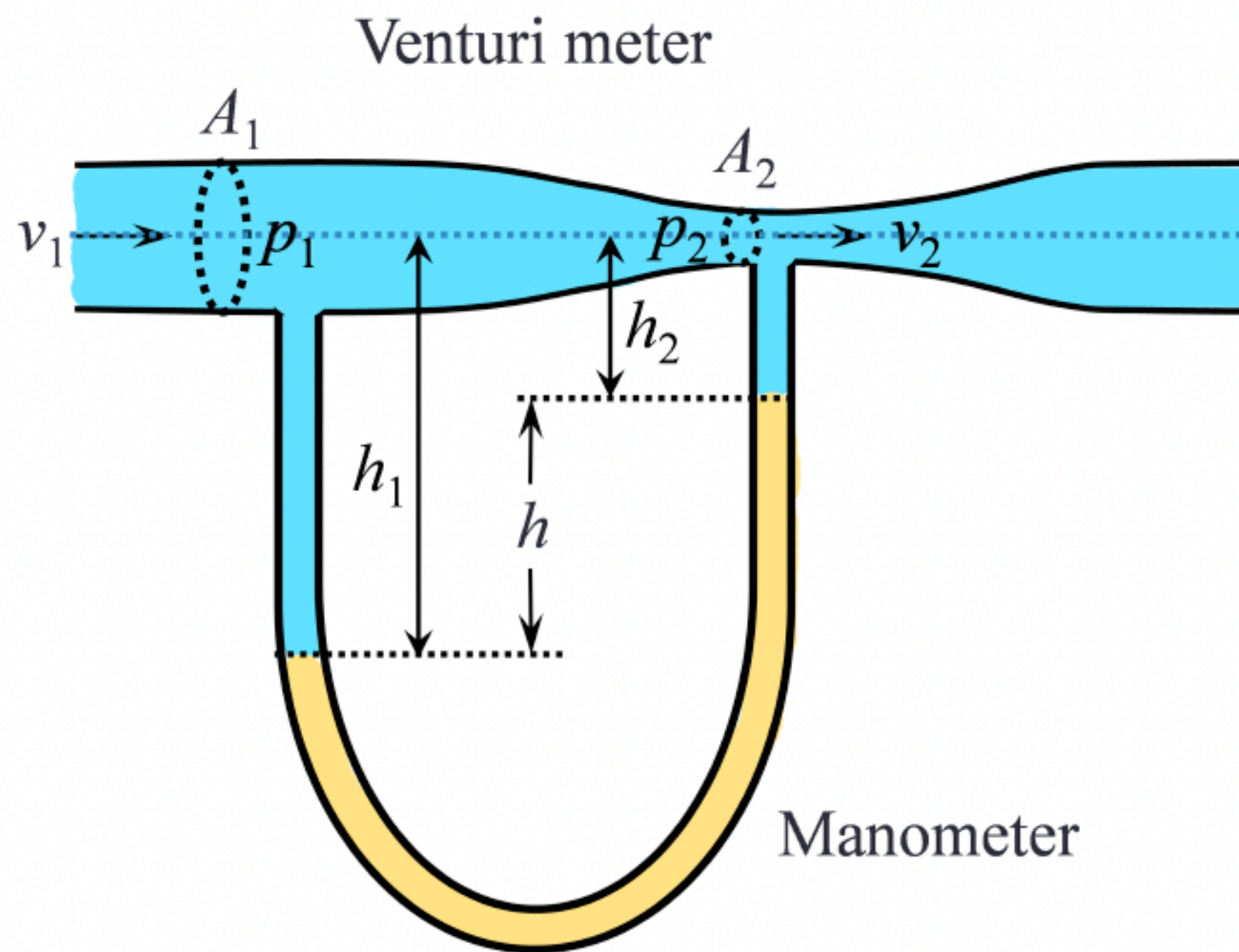
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Recap:

- $1 \text{ atm} = 10^5 \text{ N/m}^2 = 10^5 \text{ Pa}$
- Buoyancy $F_B = \rho_f g V$, where
 - F_B : buoyancy force
 - ρ_f : density of fluid
 - V : volume of the object in the fluid
- Buoyancy is always upwards
- Bernoulli's equation: $\frac{1}{2}\rho v^2 + \rho gh + P = \text{const.}$ ↙ pressure

6. As shown below, the Venturi meter is connected to a U-shaped manometer. Consider an ideal fluid of density ρ_1 flowing through the horizontal pipe. The fluid at the portion of the pipe with a larger cross-section area A_1 has a flow speed v_1 and a measured pressure of p_1 . The fluid at the portion of the pipe with a narrower cross-section area A_2 has a flow speed v_2 and a measured pressure of p_2 . There is a steady flow in the horizontal pipe and the height difference of the fluid in the manometer remains constant at $h = h_1 - h_2$. **(10 pts)**





(a) Write down the continuity equation for the horizontal pipe.

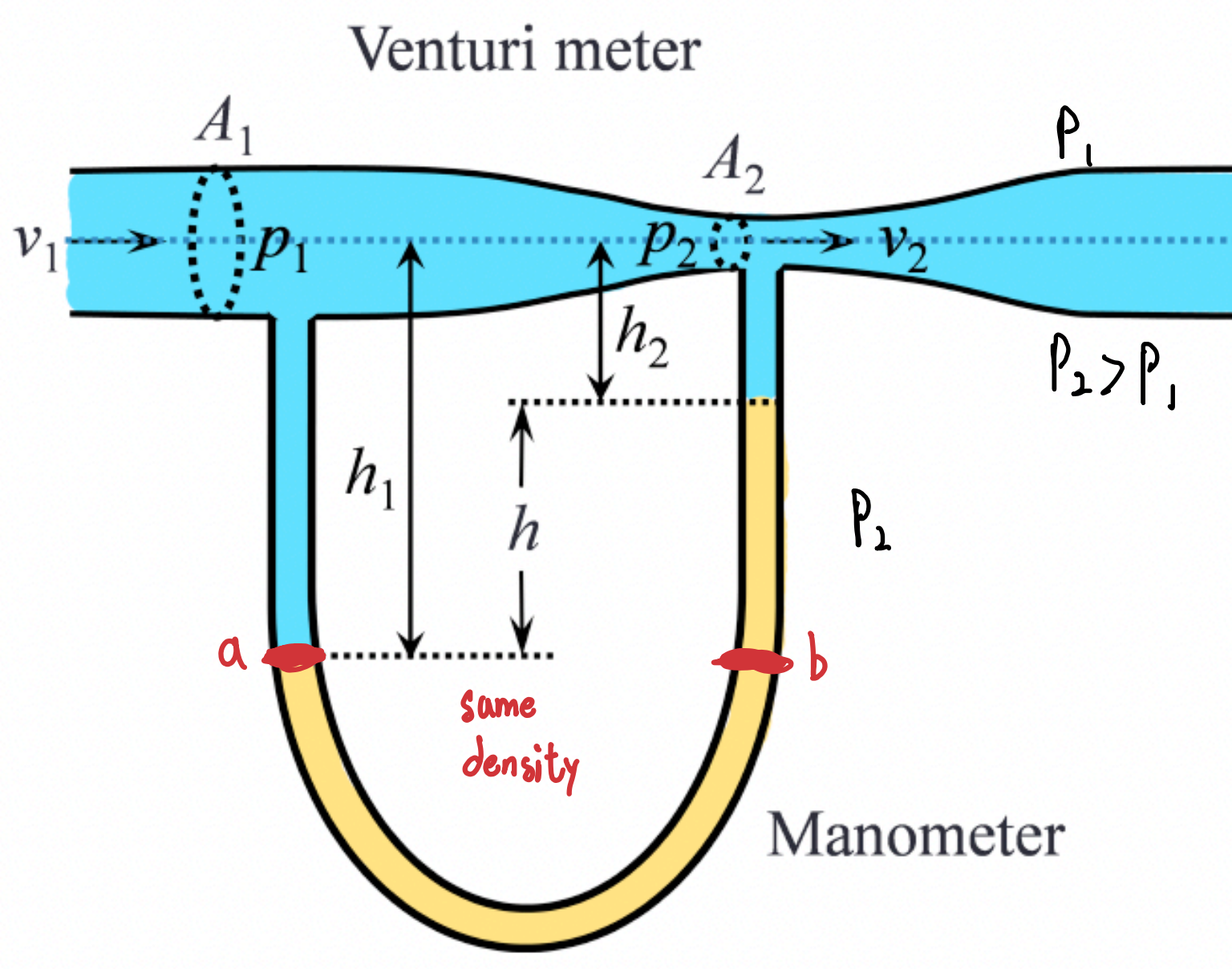
$$V_1 A_1 = V_2 A_2$$

(b) Use Bernoulli's equation to show $v_1 = \sqrt{\frac{2(p_1 - p_2)}{\rho_1[(A_1/A_2)^2 - 1]}}$

$$\left\{ \begin{aligned} \frac{1}{2} \rho_1 v_1^2 + \rho_1 g h + p_1 &= \frac{1}{2} \rho_1 v_2^2 + \rho_1 g h + p_2 \\ v_2 &= \frac{v_1 A_1}{A_2} \end{aligned} \right.$$

$$\frac{1}{2} \rho_1 v_1^2 + p_1 = \frac{1}{2} \rho_1 \left(\frac{A_1}{A_2} v_1 \right)^2 + p_2$$

$$v_1 = \sqrt{\quad}$$



- hydrostatic pressure

- continuity eq.

(c) Consider two different fluids of density ρ_1 (fluid in the horizontal pipe) and ρ_2 (fluid in the U-shaped manometer), where $\rho_1 < \rho_2$. Determine the flow speed v_1 in terms of ρ_1 , ρ_2 , A_1 , A_2 and h , using part b. **(4 pts)**

$$V_1 = \sqrt{\frac{2(P_1 - P_2)}{\rho_1 \left[\left(\frac{A_1}{A_2} \right)^2 - 1 \right]}}$$

$$P_a = P_1 + \rho_1 g h_1$$

$$P_b = P_2 + \rho_1 g h_2 + \rho_2 g h$$

$$P_1 + \rho_1 g h_1 = P_2 + \rho_1 g h_2 + \rho_2 g h$$

$$P_1 + \rho_1 g h = P_2 + \rho_2 g h$$

$$P_1 - P_2 = (\rho_2 - \rho_1) g h$$

$$V_1 = \sqrt{\frac{2(\rho_2 - \rho_1) g h}{\rho_1 \left[\left(\frac{A_1}{A_2} \right)^2 - 1 \right]}}$$

5. In the figure below, the fresh water behind a reservoir dam has depth $D = 12$ m. A horizontal pipe passes through the dam at depth $d = 6.0$ m. The inner and outer parts of the pipe have cross-section areas of $S_1 = 18.8\text{cm}^2$ and $S_2 = 12.5\text{cm}^2$. A plug secures the pipe opening. **(10 pts)**

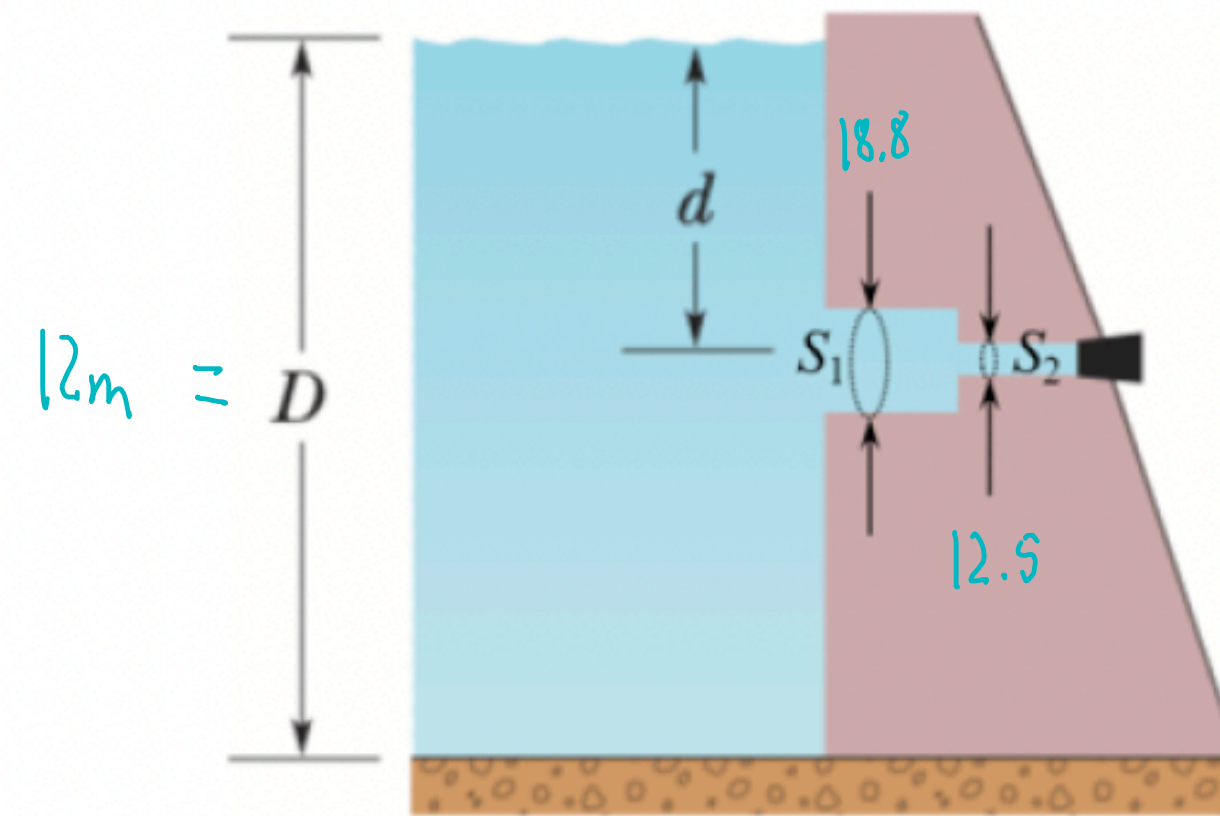
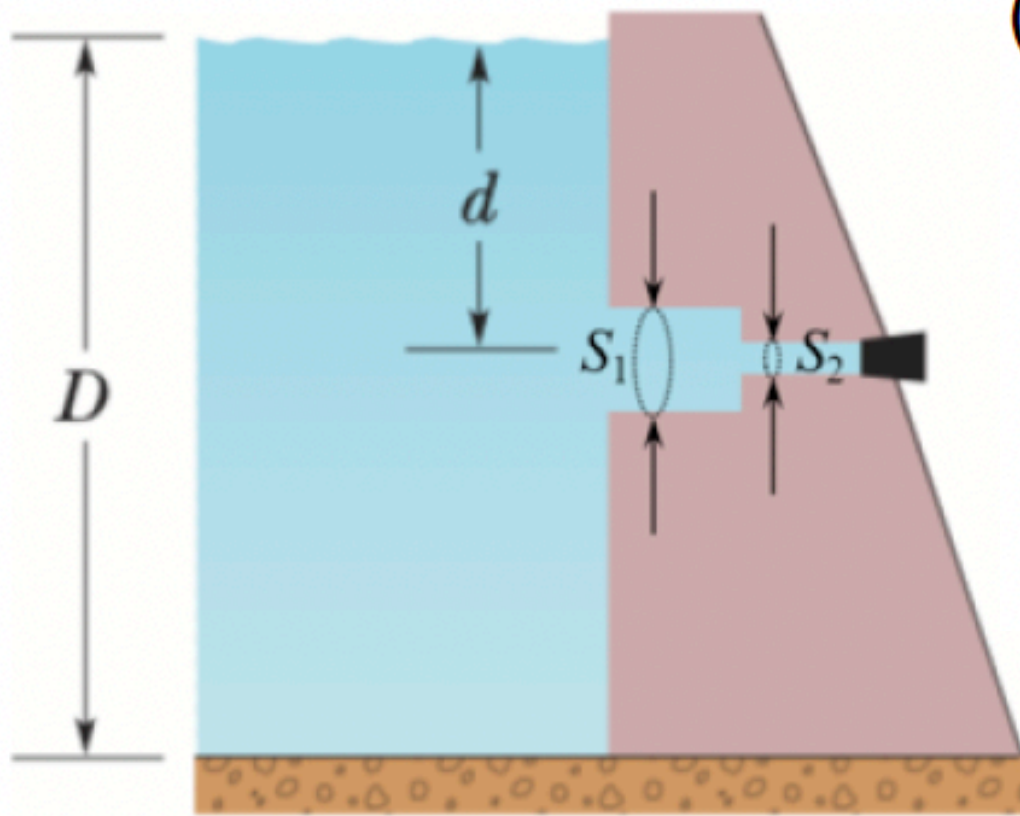


Fig. 4



(a) Find the values of gauge pressure in the horizontal pipe?

(2 pts)

(Hint: The two diameters of the horizontal pipe are much smaller than d and D . The gauge pressure of water at the cross-section of S_1 approximately equals to that at the cross-section of S_2 .) (The gauge pressure is defined as the difference between an absolute pressure (P) and the atmospheric pressure (P_{atm}).)

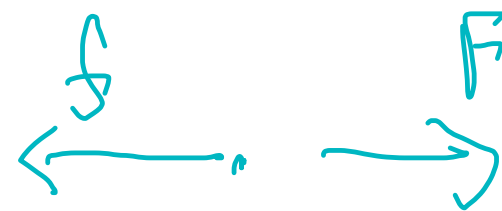
$$P = P_{atm} + \rho g d$$

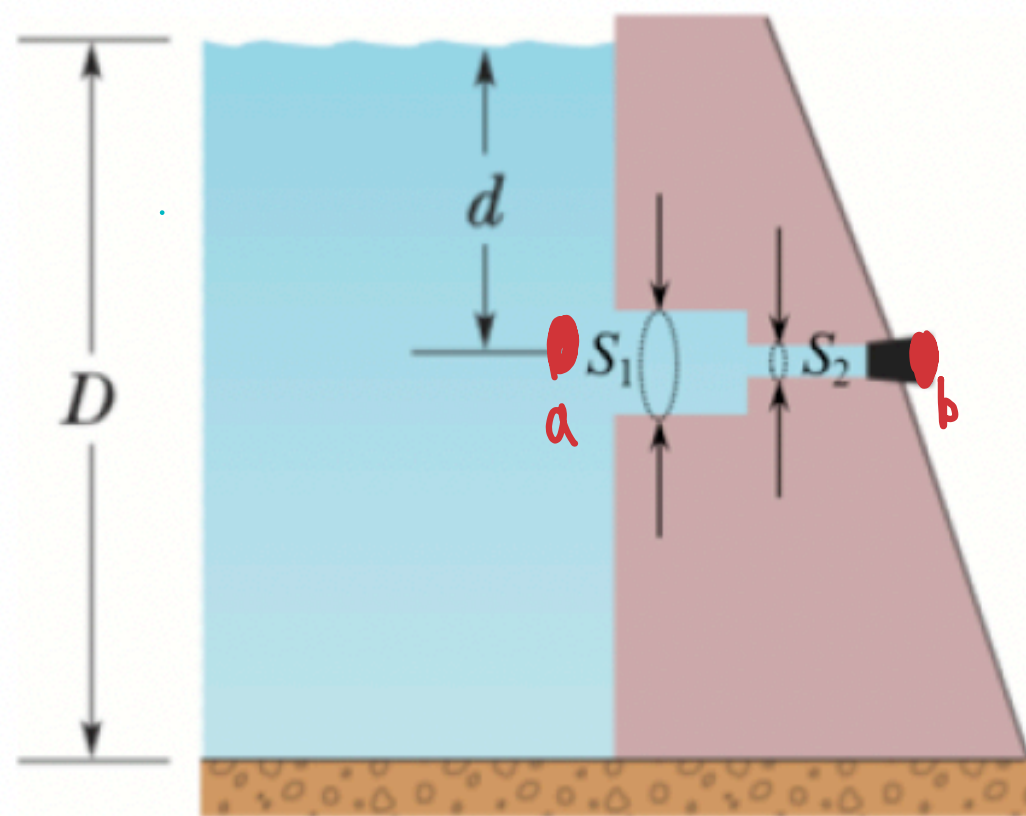
$$P_g = P - P_{atm} = \rho g d = 5.9 \times 10^4 \text{ Pa}$$

(b) Find the magnitude of the frictional force between plug and pipe wall.

(2 pts)

$$f = F = P_g S_2$$





(c) When the plug is removed, what is the water volume that exits the pipe in 1.0 hour? **(3 pts)**

$$\frac{1}{2} \rho v^2 + \rho g h + P_{atm} = 0 + \rho g h + P_{atm} + \rho g d$$

(not moving)

$$v = \sqrt{2gd}$$

$$V = v \cdot S_2 \cdot T$$

$$= 49 \text{ m}^3$$

(d) What is the speed of water flowing through the cross-section of S_1 ?

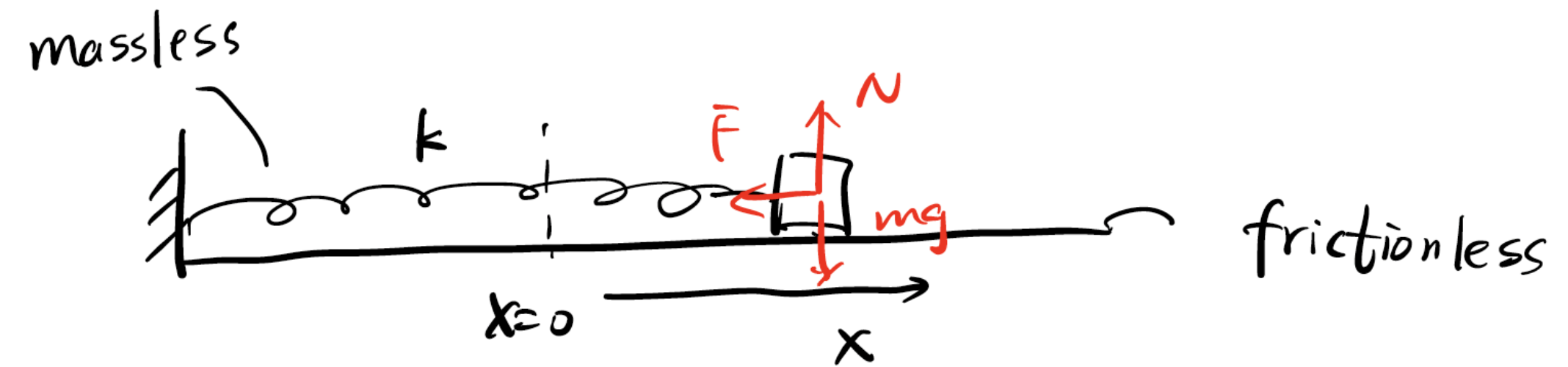
(3 pts)

$$v_1 S_1 = v_2 S_2$$

$$v_1 = \frac{S_2}{S_1} v_2 = 7.3 \text{ m/s}$$

Simple Harmonic Motion (SHM)

- Hooke's Law: $\vec{F} = -k\vec{x}$
- $\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$
- $x = A \cos(\omega t + \phi)$, where
 - $\omega = \sqrt{k/m}$
 - $T = 2\pi\sqrt{\frac{m}{k}} \approx \frac{w}{2\pi}$



$$F = -kx$$

$$= ma$$

$$= m \frac{d^2x}{dt^2}$$

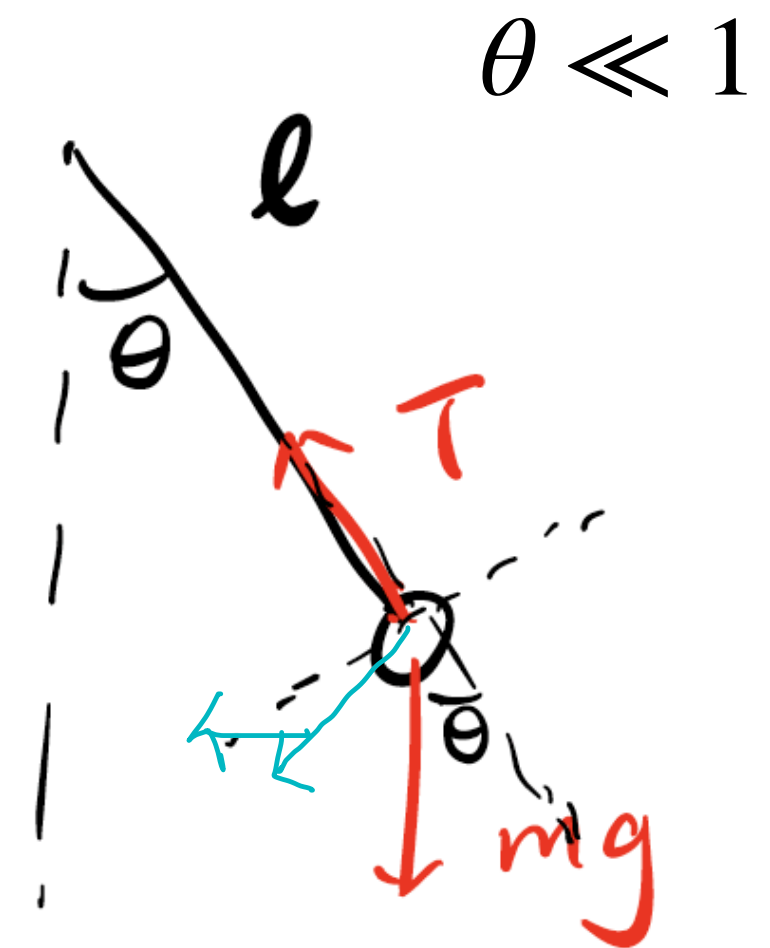
$$m \frac{d^2x}{dt^2} + kx = 0$$

$$\omega = \sqrt{\frac{k}{m}}$$

Simple Pendulum

- $\frac{d^2\theta}{dt^2} + \frac{g}{l}\theta = 0$
- $\theta = A \cos(\omega t + \phi)$, where
 - $\omega = \sqrt{\frac{g}{l}}$
 - $T = 2\pi\sqrt{\frac{l}{g}}$

$$\begin{aligned}
 F &\approx mg \sin \theta \\
 &= ma \\
 &\approx -m \alpha \cdot l \\
 &= -m \frac{d^2\theta}{dt^2} \cdot l
 \end{aligned}$$



small angle approx. $\sin \theta \approx \theta$
 θ in radian ($\theta \ll 1$)

$$mg \theta + m \frac{d^2\theta}{dt^2} \cdot l = 0$$

$$\frac{d^2\theta}{dt^2} + \frac{g}{l}\theta = 0$$

Fluid in Oscillation